

FORMULA + TUTORIAL

DAYLIGHT DAY CREAM

A practical, data-locked 100 g home R&D prototype

Designed for fast visible hydration, smoother feel and healthy-looking glow - with a realistic long-term path for uneven tone.

THE DESIGN BRIEF

4-DAY COSMETIC WIN

Hydrated + fresh + smooth + comfortably glowy

pH 5.2-5.6 | fragrance-free | 1.00% preservative blend | no steroid / hydroquinone

REALISTIC EXPECTATION

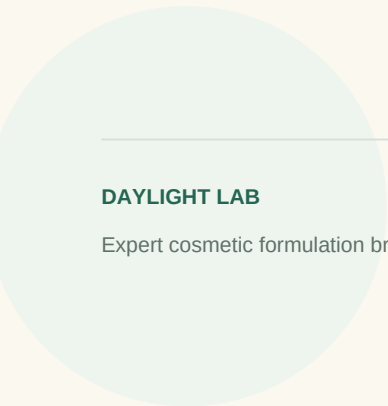
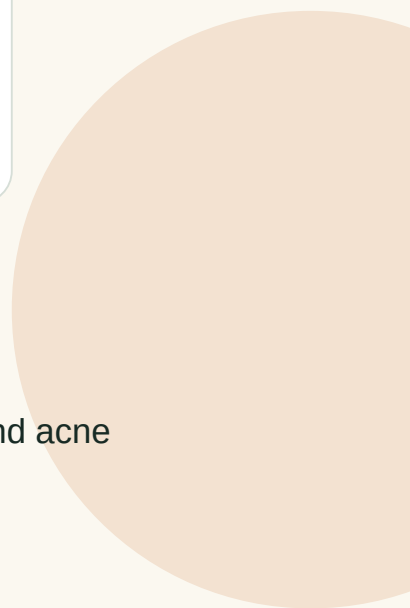
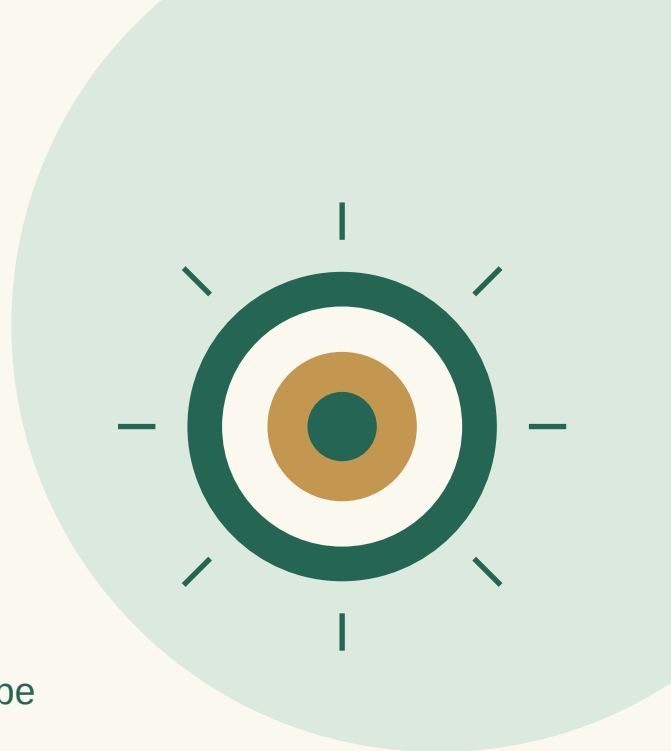
Surface hydration can look better in days. Established pigmentation and acne biology need weeks, consistency and daytime sunscreen.

Prepared as an educational R&D document
for a single small, personal-use prototype.

DAYLIGHT LAB

Expert cosmetic formulation brief | India sourcing orientation | 01 Oct 2026

VERSION 1.0



What can realistically change - and when

FOUR-DAY FOCUS

Humectants, light emollients and a smooth emulsion can quickly improve the way skin feels and reflects light. They cannot safely erase melanin or cure acne in four days.

DAY 1-4

More hydrated, smoother and fresher-looking; light reflects more evenly from a moisturised surface. Not pigment removal or acne cure.

1 WEEK

Less tightness/flaking and more comfortable skin are realistic if the formula suits the skin. Breakouts may be unchanged.

2-4 WEEKS

A gradual improvement in overall tone appearance and post-blemish look may begin. Results differ by skin, routine and sun exposure.

6-12 WEEKS

Meaningful change in uneven pigmentation is a longer-term target. Daily broad-spectrum sunscreen is essential; melasma or persistent acne needs a dermatologist.

USE ROUTINE

Morning: apply a small amount to clean dry skin, then use a broad-spectrum SPF 30+ sunscreen. For pigment-prone skin, a tinted iron-oxide sunscreen is especially useful [3].

This cream is not SPF-tested and must never be represented as sun protection.

The locked 100 g formula

Fragrance-free master formula. Percent equals grams for a 100 g batch. Total = 100.00 g.

PHASE A | WATER PHASE

HEAT TO 70-75 C

INGREDIENT	WHAT IT DOES / WHY	%	GRAMS
Purified / deionized water	Main carrier for a clean, light emulsion.	43.69	43.69
Glycerin	Immediate humectant; pulls water into the outer skin.	4.00	4.00
1,3-Propanediol	Humectant and slip aid; helps a less tacky feel.	3.00	3.00
Betaine (trimethylglycine)	Osmolyte humectant for comfortable hydration.	1.00	1.00
Allantoin	Skin-conditioning powder; supports a smoother feel.	0.20	0.20
Disodium EDTA	Chelator; supports preservative and formula stability.	0.10	0.10
Xanthan gum (transparent grade)	Low-dose thickener; helps keep the emulsion stable.	0.15	0.15

PHASE B | OIL PHASE

HEAT TO 70-75 C

INGREDIENT	WHAT IT DOES / WHY	%	GRAMS
Caprylic/Capric Triglyceride	Light emollient for softness without a waxy film.	4.00	4.00
Dicaprylyl Carbonate	Fast-spreading dry emollient for quick absorption.	3.00	3.00
Squalane	Stable, skin-compatible emollient; supports softness and barrier feel.	2.00	2.00
Olivem 1000 (Cetearyl Olivat and Sorbitan Olivat)	Primary emulsifier; binds water and oils into cream.	4.50	4.50
Cetearyl alcohol	Co-emulsifier; gives cushion, body and stability.	0.70	0.70
Mixed tocopherol (min. 70% active)	Oil-phase antioxidant; protects the emollient phase.	0.20	0.20

Texture intention: light lotion-cream, quick-spreading, low-wax finish. The gel network + Olivem 1000 + cetearyl alcohol provide stability without a heavy feel.

Cool-down actives, preservation & formula logic

PHASE C | COOL-DOWN ACTIVES + PRESERVATION

ADD BELOW 40 C

INGREDIENT	WHAT IT DOES / WHY	%	GRAMS
Purified / deionized water	Cool-down carrier for dissolved actives.	25.00	25.00
Niacinamide	Brightening/barrier-support active; selected at a studied 4% level.	4.00	4.00
N-Acetyl Glucosamine (NAG)	Tone-evenness support; pairs well with niacinamide.	2.00	2.00
D-Panthenol (min. 98% powder)	Humectant/barrier-support active for skin comfort.	1.00	1.00
Zinc PCA	Blemish-prone/oil-balance support; not an acne medicine.	0.30	0.30
Trisodium citrate dihydrate	Part of a gentle citrate buffer for the target pH.	0.11	0.11
Citric acid, anhydrous	Part of the citrate buffer; supports pH control.	0.05	0.05
Phenoxyethanol (and) Ethylhexylglycerin	Broad preservative blend (e.g., Euxyl PE 9010).	1.00	1.00

FAST LOOK

4.00% glycerin + 3.00% propanediol + 1.00% betaine and light esters support quick hydration, slip and surface glow.

LONGER PATH

4.00% niacinamide + 2.00% NAG match concentrations used together in published pigmentation-appearance research [1].

SAFETY CORE

Target pH 5.2-5.6; 1.00% preservative blend; airless pack. Challenge testing is required before any sale [4,5].

SENSORY OPTION - NOT THE LOCKED FORMULA

Keep fragrance-free for the safest first prototype. If a scent is required, use 0.05% IFRA-compliant cosmetic fragrance and reduce total water by 0.05% (water becomes 68.64 g); patch test. Do not use essential oils as a shortcut.

Manufacturing tutorial - small, careful, repeatable

Required: 0.01 g scale, calibrated pH meter, two heat-safe beakers, thermometer, silicone spatula, stick blender / mini homogeniser and opaque airless pump bottle.

1

01 SANITIZE

Clean bench, tools and pack. Spray 70% isopropyl alcohol; allow to air-dry fully. Wear clean gloves. Use only purified/deionized water - never tap water.

2

02 PHASE A

Weigh the 43.69 g A-water. Pre-slurry 0.15 g xanthan into the 4.00 g glycerin. Add it to water with propanediol, betaine, EDTA and allantoin; stir until uniform.

3

03 PHASE B

In the second beaker weigh CCT, dicaprylyl carbonate, squalane, Olivem 1000, cetearyl alcohol and tocopherol. Heat A and B separately to 70-75 C. Hold until B is fully melted and A is uniform (about 15-20 min).

4

04 EMULSIFY

With A and B within 3 C, slowly pour B into A. Blend 60-90 seconds in short pulses to minimise air. Then stir gently for 8-10 min. Do not whip.

5

05 COOL

Stir gently while cooling. At 40 C or below, make sure phase C is a clear solution: dissolve citrate salts, niacinamide, NAG, panthenol and Zinc PCA in its 25.00 g water (gentle 35-40 C is allowed).

6

06 FINISH

Add phase C and the preservative blend below 40 C; mix 5 min. Check pH at 25 C after 10 min rest: target 5.2-5.6 (aim 5.4). Fill only when below 30 C.

CRITICAL CONTROL

Do not add the cool-down actives or preservative to a hot batch. If phase C is not fully dissolved, do not fill a gritty / unstable product.



pH, fill, stability & the small details that matter

MEASURED PH - DO NOT GUESS

The fixed citrate buffer is designed around pH 5.2-5.6 (aim 5.4 at 25 C), but raw-material lots change pH. Calibrate a meter with pH 4.01 and 7.00 buffers. If the first pilot is >5.6, titrate a 10% citric-acid solution one drop at a time, wait 10 min, recheck and record the amount; subtract the same mass from water in the next batch. If <5.2, do not use an unmeasured alkali - reformulate with a trained formulator.

FILL + STORE

Fill below 30 C into an opaque airless pump. Avoid jars: fingers raise contamination risk. Store sealed at 15-25 C, dry and away from sun. Because this home batch has no challenge test, make only 100 g and use within 4 weeks after opening.

HOME STABILITY SCREEN

Retain samples at room temperature, 4 C and 40 C for 4 weeks. Inspect weekly: separation, grains, odour, colour, pump function and pH. A pH shift >0.3, separation or odour change = fail. This is screening, not proof of shelf life.

FORMULATION TRICKS THAT EARN THEIR PLACE

Keep phase temperatures aligned

A/B within 3 C avoids shocky, weak emulsions.

Pre-slurry the xanthan

Glycerin wetting prevents stubborn gum fish-eyes.

Use short blending pulses

Enough shear for small droplets; less trapped air and foam.

Reserve cool-down for actives

It protects panthenol and keeps the sensory profile clean.

Choose the airless pack

Better user experience and lower in-use contamination than a jar.

Make small batches

No challenge test means no claimed shelf life or sale.

STOP / DISCARD IF

Mould, gas, unusual odour, colour change, new separation, watery thinning, gritty crystals, package swelling or unexpected irritation appear. Preservation must be proven by a suitable challenge test before commercial distribution [5].

India shopping list & practical prototype cost

Buy cosmetic raw materials with a current COA, SDS, batch number and expiry. Prices are approximate online orientation only, checked 01 Oct 2026; shipping, GST and pack size change them.

Niacinamide	cosmetic/Ph.Eur grade, >=99%	100 g	ASES Chemical Works lists 99.5% / 100 g at about Rs 240; confirm COA and stock.
Zinc PCA	cosmetic grade, water soluble	30 g	Cosmesi Global lists cosmetic Zinc PCA; 100 g quoted around Rs 750; confirm COA and stock.
NAG + D-Panthenol	NAG >=98%; D-Panthenol >=98%	25 g each	Buy only with batch COA/SDS from a cosmetic raw-material supplier.
Olivem 1000 + Cetearyl alcohol	correct INCI / cosmetic grade	100 g each	Euroasia Cosmetics and other Indian cosmetic raw suppliers list cream emulsifiers; verify INCI.
CCT, Dicaprylyl Carbonate, Squalane	cosmetic grade	100 mL each	Ask for oxidation/identity documents and date of manufacture.
Euxyl PE 9010 or verified equivalent	Phenoxyethanol (and) Ethylhexylglycerin	30 g	Use the supplier-prescribed concentration and check the exact INCI; do not substitute a medicine.
Water phase / support materials	DI water, glycerin USP, 1,3-propanediol, betaine, allantoin, EDTA, xanthan, citrate salts	25-100 g	Cosmetic/lab grade, with COA. Do not use unknown food or pharmacy powders as substitutes.

BATCH COST ORIENTATION

BATCH	RAW MATERIALS	AIRLESS PACK + LABEL	ESTIMATED TOTAL
100 g	Rs 65-110	Rs 40-90	Rs 105-200
500 g	Rs 280-450	Rs 200-450	Rs 480-900
1 kg	Rs 520-800	Rs 400-900	Rs 920-1,700

IMPORTANT COST NOTE

Minimum purchase for the first kit can be roughly Rs 3,500-6,000 before tools/shipping because actives and oils are sold in small packs. Do not cut cost by omitting preservative, pH measurement or clean packaging.

Safety, boundaries & evidence

PATCH TEST

Apply a rice-grain amount to the inner forearm or behind the ear for 2-3 days. Stop for burning, swelling, rash or persistent itching. Do not apply to broken / infected skin or the eye area.

PRESERVATION

A 1.00% phenoxyethanol / ethylhexylglycerin blend is a sensible starting system, not proof. The actual product + pack needs preservative efficacy (challenge) testing before sale [4,5].

DO NOT IMPROVISE

Do not tip raw AHA/BHA, L-ascorbic acid, benzoyl peroxide, retinoids, essential oils or drug powders into this cream. They change pH, stability, irritation risk and preservation.

DAYTIME RULE

Use this moisturiser under a broad-spectrum SPF 30+ sunscreen every day. For pigmentation, choose a tinted iron-oxide sunscreen where suitable [3]. This cream has no UV filters.

'NATURAL' IS NOT AUTOMATICALLY SAFER

Natural fragrance, citrus oil and unpreserved plant waters can sensitise skin or introduce microbes. Safety comes from identity, purity, concentration, preservation, testing and correct use - not a marketing word.

WHEN TO SEE A DERMATOLOGIST

Painful/cystic acne, scarring, sudden pigmentation, melasma, eczema, pregnancy-specific questions or a reaction need medical advice. This cosmetic prototype does not replace diagnosis or treatment.

KEY SOURCES

[1] Bissett DL et al. J Cosmet Dermatol. 2007;6:20-26. 2% NAG, with greater effect in a 2% NAG + 4% niacinamide study after 8 weeks. PMID 17348991. <https://pubmed.ncbi.nlm.nih.gov/17348991/>

[2] Hakozaki T et al. Br J Dermatol. 2002;147:20-31. Topical niacinamide and melanosome transfer / pigmentation appearance. DOI 10.1046/j.1365-2133.2002.04834.x

[3] American Academy of Dermatology. Dark spots: tinted iron-oxide sunscreen, broad spectrum, water resistant, SPF 30+; reapply every two hours. <https://www.aad.org/public/everyday-care/skin-care-secrets/routine/fade-dark-spots>

[4] SCCS/1575/16 (2016). Phenoxyethanol considered safe as a preservative in cosmetics at a maximum 1.0%. https://ec.europa.eu/health/scientific_committees/consumer_safety/docs/sccs_o_195.pdf

[5] FDA Cosmetic Microbiological Safety (2012). A preservative efficacy / challenge test is used to confirm preservation for the actual formula and pack. https://downloads.regulations.gov/FDA-2011-N-0770-0009/attachment_1.pdf

DATA LOCK CHECK | Both deliverables are generated from one formula list: total 100.00 g, pH 5.2-5.6, A/B 70-75 C, C below 40 C, fill below 30 C, preservative blend 1.00%.